4. $\left(\frac{2}{7},\frac{3}{7},\frac{6}{7}\right)$, $\vec{r}=\left(5\hat{i}+6\hat{j}+7\hat{k}\right)+t\left(2\hat{i}+3\hat{j}+6\hat{k}\right)$ or

$\vec{r}=\left(7\hat{i}+9\hat{j}+13\hat{k}\right)+t\left(2\hat{i}+3\hat{j}+6\hat{k}\right)$,

$\frac{x-5}{2}=\frac{y-6}{3}=\frac{z-7}{6}$ or $\frac{x-7}{2}=\frac{y-9}{3}=\frac{z-13}{6}$

5. (i) $0°$ (ii) $\frac{\pi}{6}$ (iii) $\frac{\pi}{2}$ 6. $\frac{\pi}{2}$ 7. $a=18, b=\frac{2}{3}$ 8. 1

Exercise 6.5

1. $\vec{r}=\left(5\hat{i}+2\hat{j}+8\hat{k}\right)+t\left(2\hat{i}+\hat{j}-2\hat{k}\right), t\in\mathbb{R}$, $\frac{x-5}{2}=\frac{y-2}{1}=\frac{z-8}{-2}$
2. $\frac{7}{\sqrt{5}}$ units 3. $\frac{9}{2}$ 4. $(6,2,1)$ 5. 2 units
6. $\frac{\sqrt{83}}{\sqrt{6}}$ units 7. $(1,6,0)$, $\frac{x-5}{-4}=\frac{y-4}{2}=\frac{z-2}{-2}$

Exercise 6.6

1. $\vec{r}\cdot\left(\frac{3\hat{i}-4\hat{j}+5\hat{k}}{5\sqrt{2}}\right)=7$ 2. $\frac{12}{13},\frac{3}{13},\frac{-4}{13}$; $\vec{r}.\left(\frac{12\hat{i}+3\hat{j}-4\hat{k}}{13}\right)=5$; 5
3. $\vec{r}\cdot\left(\hat{i}+3\hat{j}+5\hat{k}\right)=35$; $x+3y+5z=35$ 4. $\vec{r}\cdot\left(\hat{i}+\hat{j}+\hat{k}\right)=2$; $x+y+z=2$
5. x-intercept $=2$, y- intercept $=3$, z- intercept $=-4$ 6. $\frac{x}{u}+\frac{y}{v}+\frac{z}{w}=3$

Exercise 6.7

1. $\vec{r}\cdot\left(\hat{i}-2\hat{j}+4\hat{k}\right)=20$; $x-2y+4z-20=0$
2. $\vec{r}\cdot\left(3\hat{i}+4\hat{j}-5\hat{k}\right)=9$; $3x+4y-5z-9=0$
3. $\vec{r}=\left(2\hat{i}+2\hat{j}+\hat{k}\right)+s\left(-\hat{i}-4\hat{j}+2\hat{k}\right)+t\left(3\hat{i}-4\hat{j}+5\hat{k}\right)s,t\in\mathbb{R}$;

   $12x-11y-16z+14=0$
4. $\vec{r}\cdot\left(\hat{i}+10\hat{j}+7\hat{k}\right)=9$; $x+10y+7z-9=0$
5. $\vec{r}=\left(\hat{i}-\hat{j}+3\hat{k}\right)+s\left(2\hat{i}-\hat{j}+4\hat{k}\right)+t\left(\hat{i}+2\hat{j}+\hat{k}\right)s,t\in\mathbb{R}$; $9x-2y-5z+4=0$
6. $\vec{r}=\left(3\hat{i}+6\hat{j}-2\hat{k}\right)+s\left(-4\hat{i}-8\hat{j}+8\hat{k}\right)+t\left(3\hat{i}-2\hat{j}\right)s,t\in\mathbb{R}$; $\vec{r}\cdot\left(2\hat{i}+3\hat{j}+4\hat{k}\right)=16$;

   $2x+3y+4z-16=0$
7. $\vec{r}\cdot\left(3\hat{i}+5\hat{j}-7\hat{k}\right)=6$; $3x+5y-7z-6=0$

Exercise 6.8

1. $\vec{r}\cdot\left(17\hat{i}-47\hat{j}-24\hat{k}\right)=-172$ 2. $x+2y-z-4=0$
3. $m=\pm\sqrt{2}$ 4. $-2, 2$, $y+z+1=0$, $y-z+1=0$

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